W2-PM5: Zenmap-Based Network Scanning

Module completed: W2-PM5 (Essential) | **Course:** Week 2 - Project Modules

1. Objective

To perform network discovery and service identification using Zenmap and Nmap in a Kali Linux virtual environment.

2. Environment

- Operating system: Kali Linux
- Scanning tool: Zenmap 7.99
- Network interface: eth0
- Scanner IP: 10.0.0.2
- Target IP: 10.0.0.1

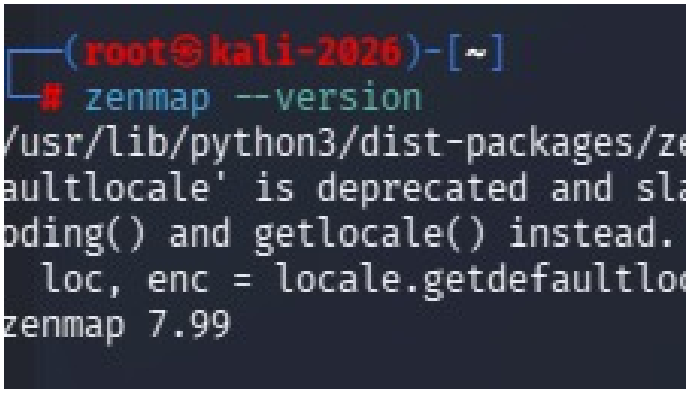


Figure 1: Zenmap version check (7.99)

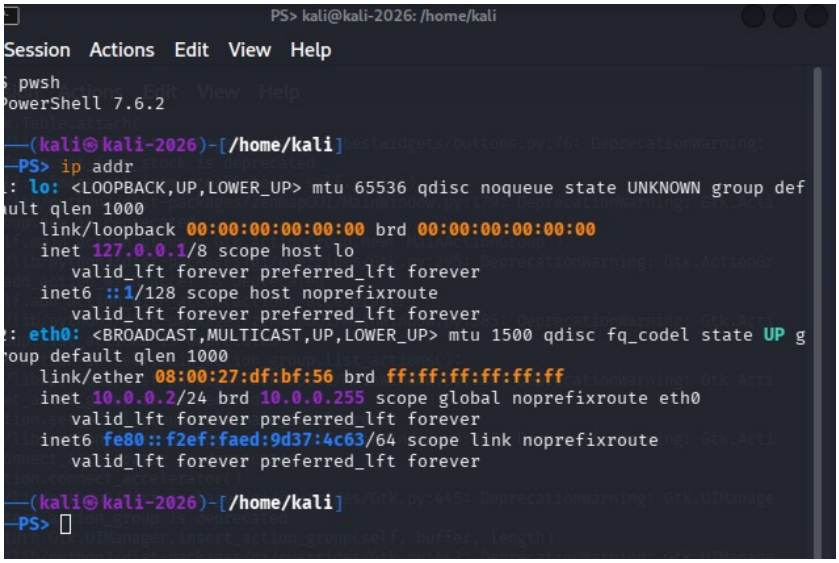


Figure 2: Scanner network configuration (eth0, 10.0.0.2/24)

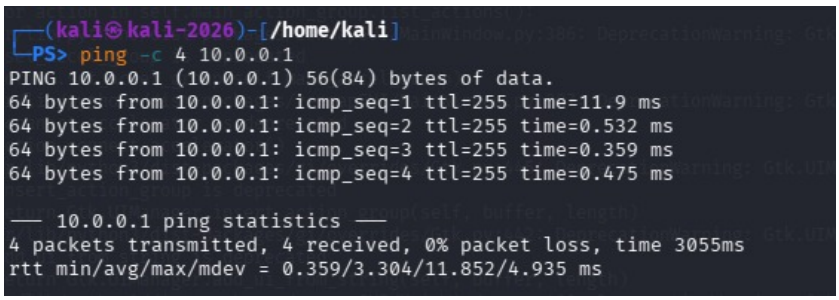


Figure 3: Connectivity test to the target with ping

3. Host Discovery

Command:

```
nmap -sn 10.0.0.1
```

Result:

- Target host was reachable.
- One host discovered.
- Response latency: 0.0017 seconds.

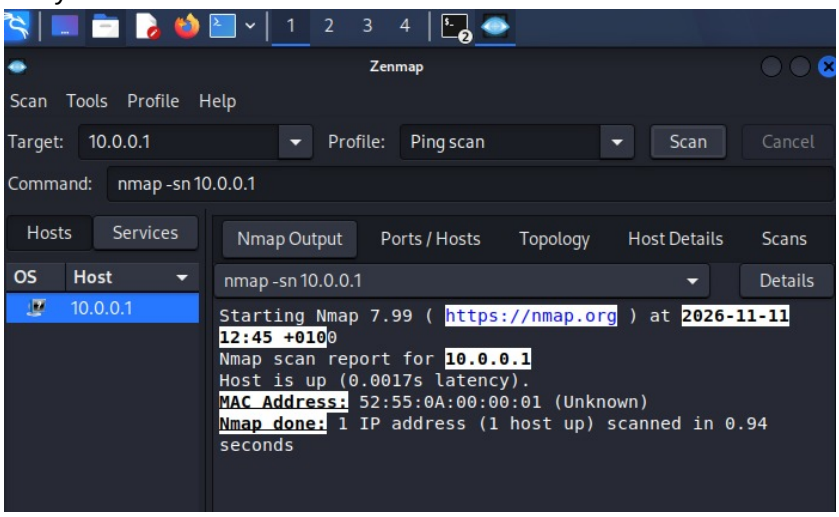


Figure 4: Ping scan of 10.0.0.1 in Zenmap

4. Port Scanning

Command:

```
nmap 10.0.0.1
```

Port	State	Service
135/tcp	Open	MSRPC
445/tcp	Open	Microsoft-DS
7070/tcp	Open	RealServer

Nmap reported 997 filtered TCP ports.

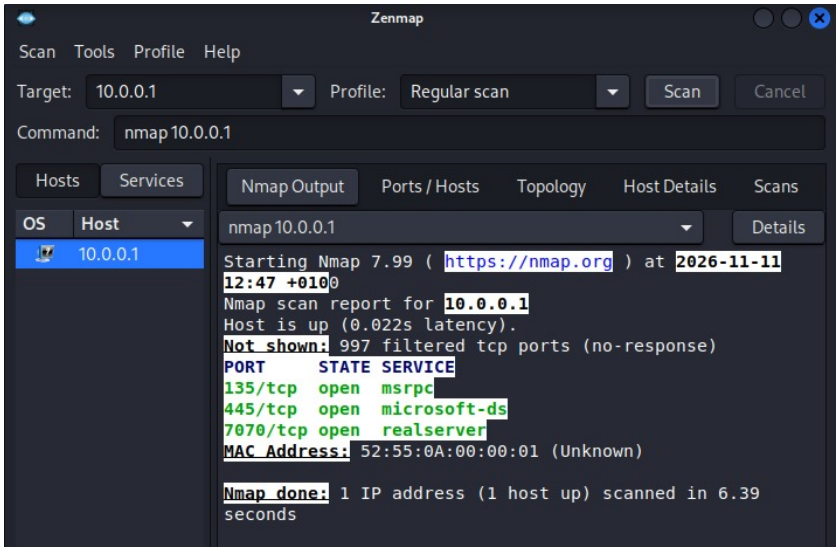


Figure 5: Regular scan of 10.0.0.1

5. Service Identification

Command:

```
nmap -sV 10.0.0.1
```

Result:

- Host was up (0.0043 s latency); scan completed in 28.89 seconds.
- 997 filtered TCP ports (no response).

Port	State	Service	Version
135/tcp	Open	msrpc	Microsoft Windows RPC
445/tcp	Open	microsoft-ds?	Not identified
7070/tcp	Open	ssl/realserver?	Not identified

- Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows.
- The question marks mean Nmap's identification of ports 445 and 7070 is uncertain.
- MAC address 52:55:0A:00:00:01 was again reported.

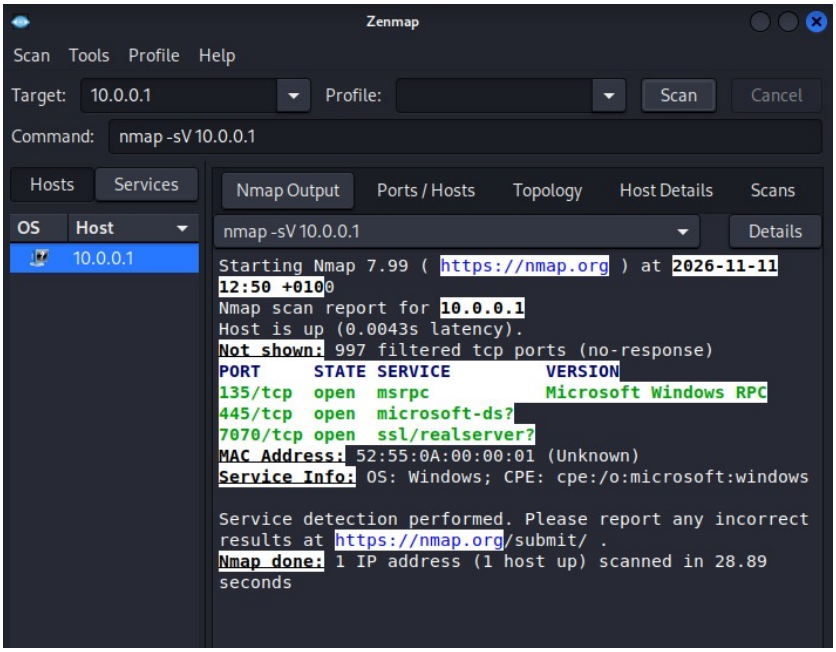


Figure 6: Service version detection scan

6. Gateway Verification

Command:

```
ip neigh show 10.0.0.1
```

The gateway MAC address matched the address reported by Zenmap: 52:55:0a:00:00:01.



Figure 7: Neighbour table entry for 10.0.0.1

7. Subnet Host Discovery

Command:

```
nmap -sn 10.0.0.0/24
```

Result:

- 256 IP addresses scanned in 74.58 seconds.
- 2 hosts found up: 10.0.0.1 and 10.0.0.2.
- 10.0.0.1 responded with MAC address 52:55:0A:00:00:01 and a latency of 2.0 seconds.
- 10.0.0.2 is the scanner's own address (confirmed by `ip addr` on eth0), so no MAC was reported for it.

The 10.0.0.1 latency was much higher than in the earlier single-host scan (0.0017 s), which is normal variation in a virtual network.

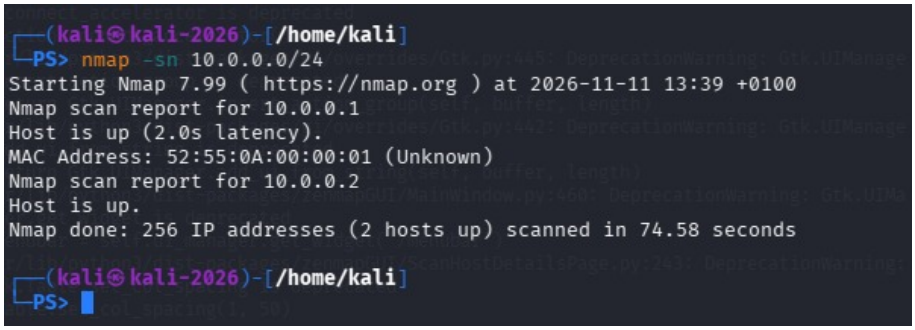


Figure 8: Subnet ping scan from the terminal

8. Network Topology

Zenmap's Topology tab, using the same subnet scan, showed:

- 10.0.0.1 as the yellow node.
- 10.0.0.2 as the green node.
- Both connected through the central localhost node.

The Hosts Viewer, Fisheye and Controls options were used to adjust the layout (Polar interpolation, Weighted layout). The topology confirms the network contains only two active hosts.

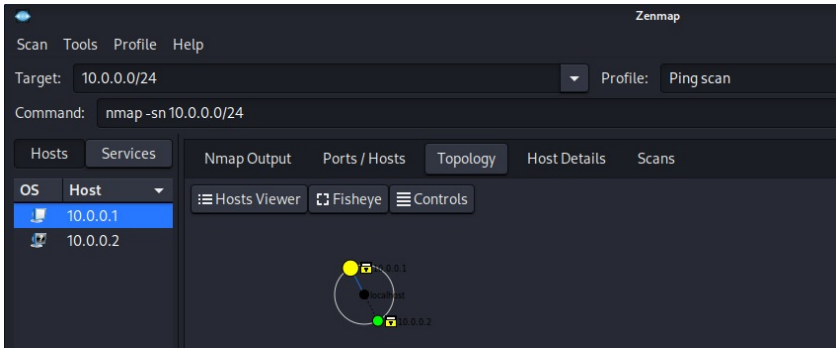


Figure 9: Topology tab (initial view)

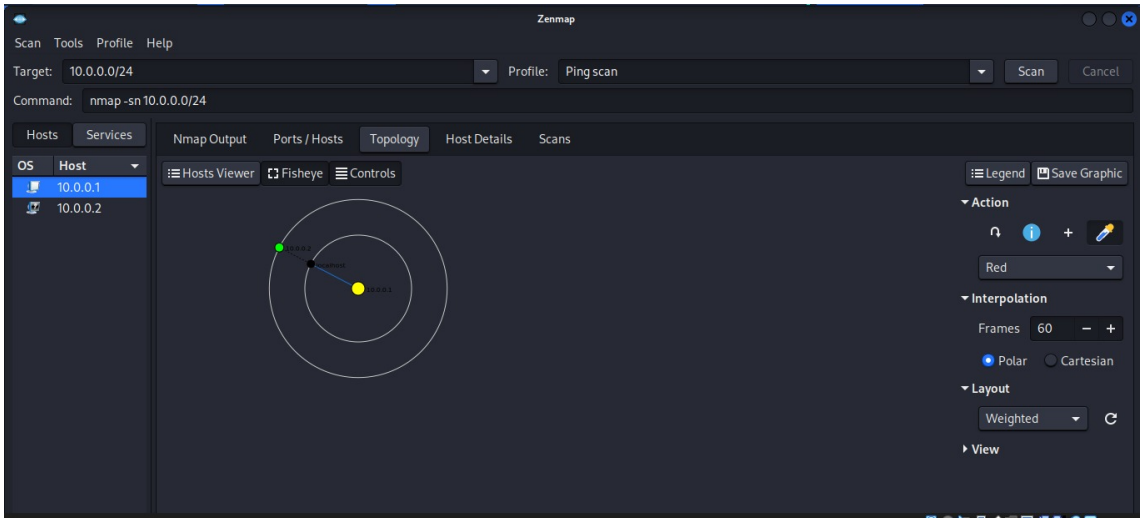


Figure 10: Topology with controls (Polar interpolation, Weighted layout)

9. Saving the Topology

The topology was exported with "Save Graphic" as `Zenmap.pdf` to the Desktop (6.3 kB). The same folder holds earlier footprinting outputs (`tesla_footprinting.json` and `.xml`).

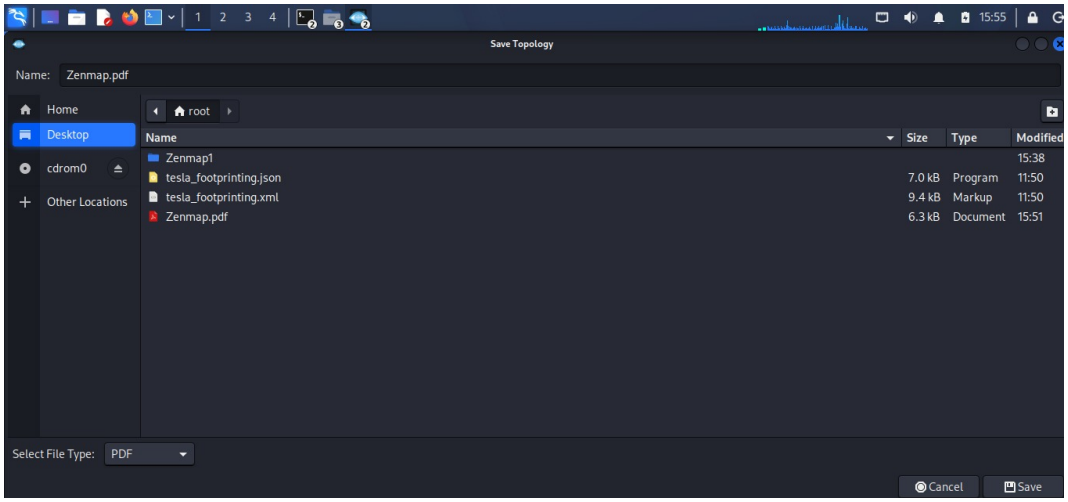


Figure 11: Save Topology dialog

10. Security Observations

The scan identified potentially sensitive network services. Ports 135 and 445 should be restricted to trusted networks where appropriate. The service on port 7070 requires further identification. A small, two-host network limits exposure, but the results do not establish that any discovered service is vulnerable or secure.

11. Conclusion

The practical demonstrated host discovery, subnet sweeping, TCP port scanning, service identification, gateway MAC verification and topology visualisation using Zenmap and Nmap.